



GS FOUNDATION 2024-25

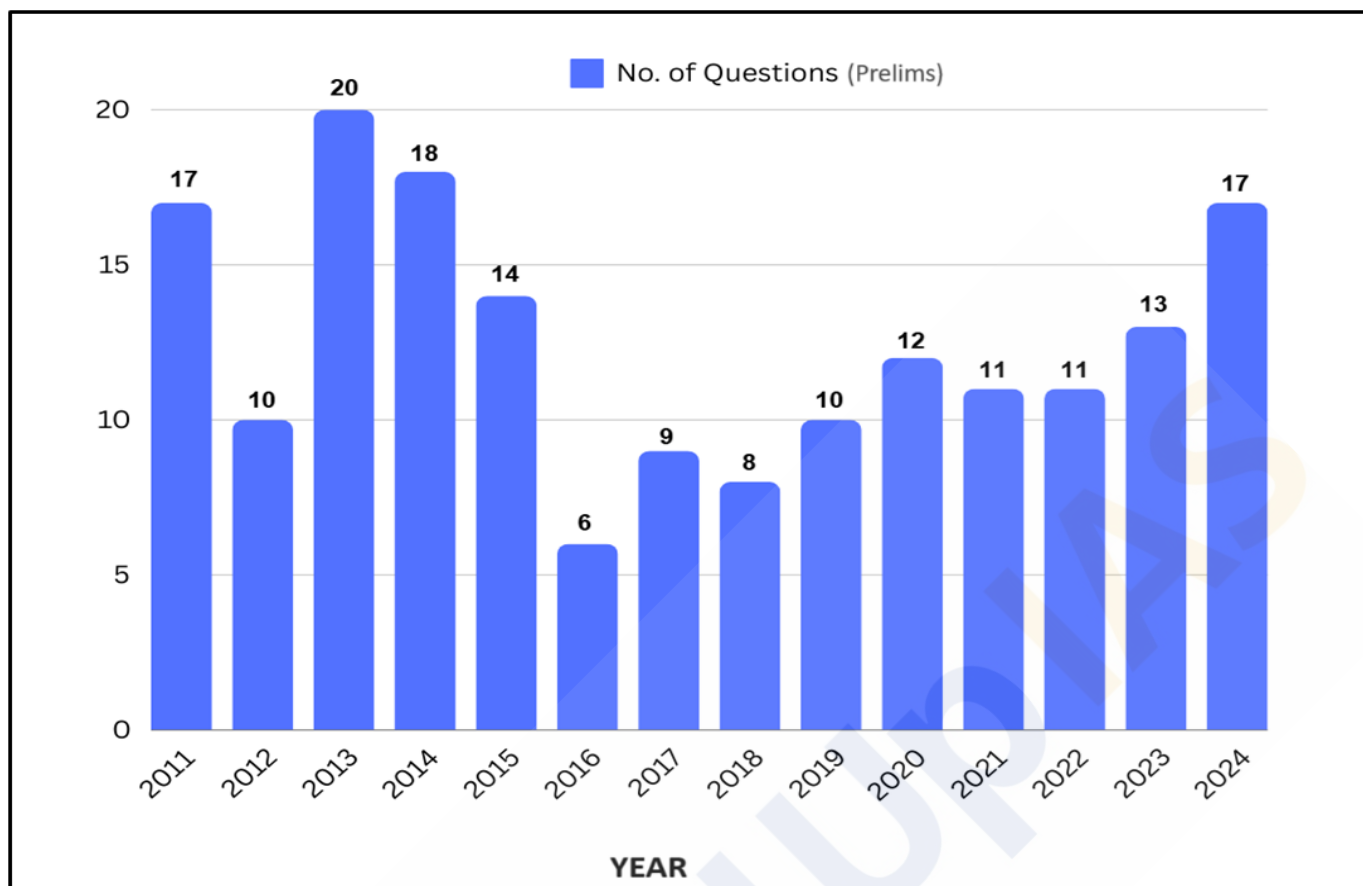
GEOGRAPHY - 17

Geomorphic Processes - I

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Year-wise Trend Analysis (2011 – 2024)



Topic-wise Trend Analysis (2011 – 2024)

Physical Geography

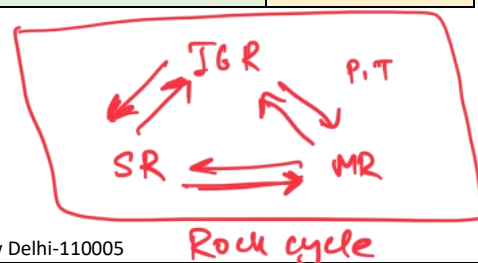
Topic	Questions
Geomorphology	9
Oceanography	5
Climatology	22
Mapping	16
Distribution of Resources	5
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Geomorphology

1. Rock, Minerals, Rock cycle
2. Movement of Earth crust → D.C.T

Indian Geography

Topic	Questions
Physiography of India	24
Drainage System of India	15
Indian Climate	7
Indian Soils	5
Natural Vegetation of India	12
Indian Agriculture	31
Mineral Resources of India	10



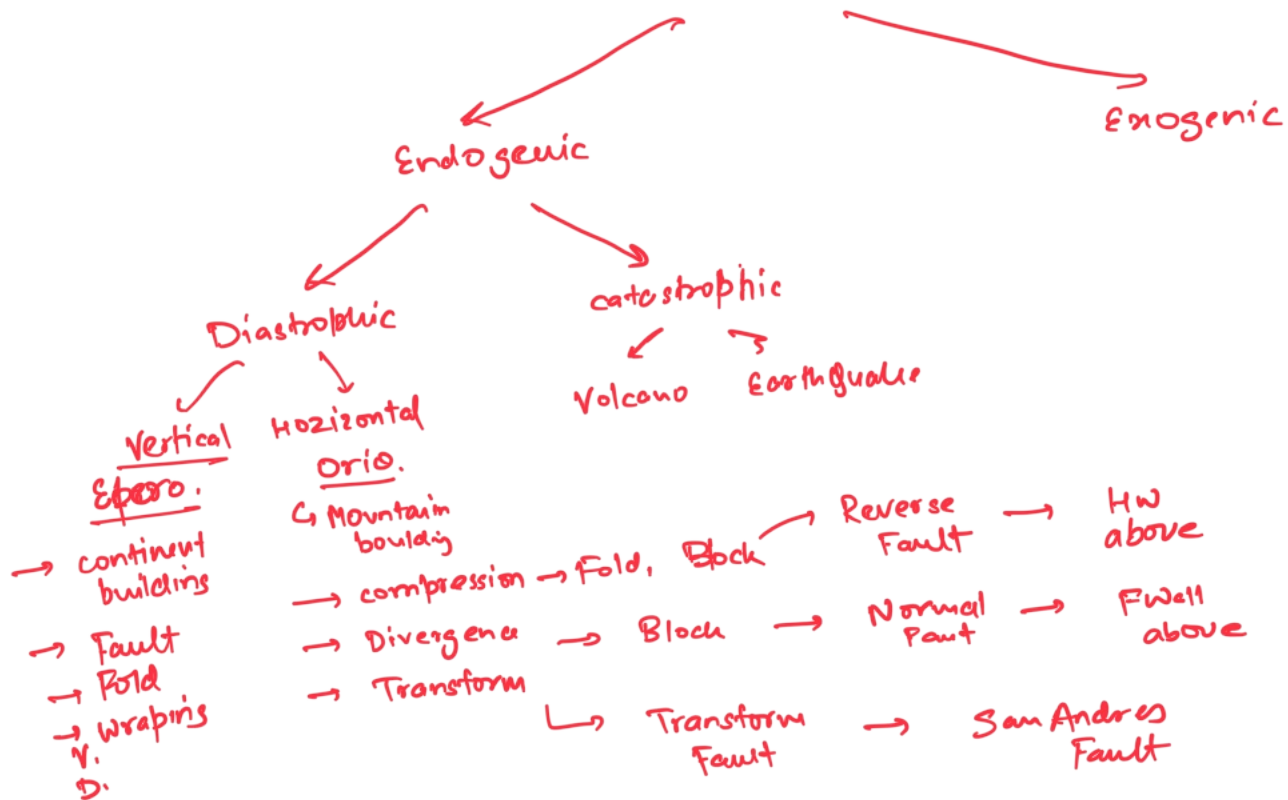
Rock cycle

3. CDT
SFS
CCC
PPT

Theory on Earth Crust

4. Processes on Earth Surfaces

Geomorphic Process



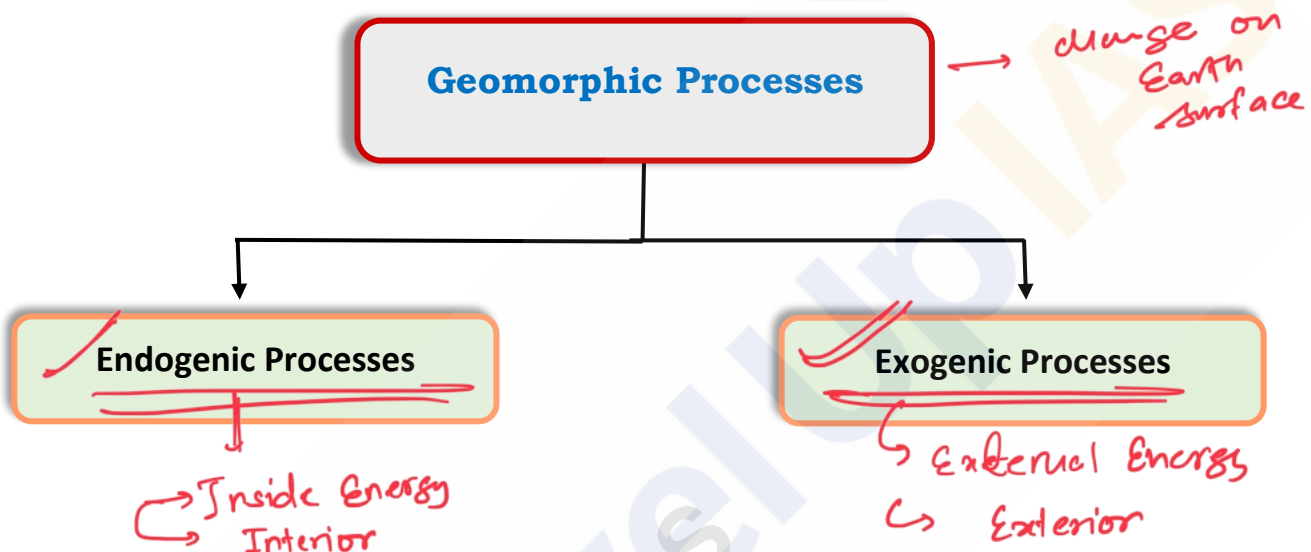
Geomorphic Processes

Introduction:

The Earth's surface is characterized by its unevenness, featuring diverse relief features like plains, plateaus, and mountains. This chapter aims to explore the reasons behind the **Earth's uneven terrain and the forces responsible** for shaping these remarkable relief features.

Geomorphic Processes:

The processes (Physical & Chemical) that bring about changes on the earth's surface are known as Geomorphic processes. These processes can be divided into two main categories: **Endogenic and Exogenic**.



Endogenic Processes:

are energized from earth interior

- The processes which ~~take place within the Earth~~ are called **Endogenic Processes**. The energy generating from **within the Earth** is the main force behind Endogenic Processes. These include:

- Radioactivity → *Disintegration of radioactive element in Earth Inside.*
- ~~Rotational Forces~~
- ~~Tidal Friction~~
- Primordial heat from the origin of the Earth.

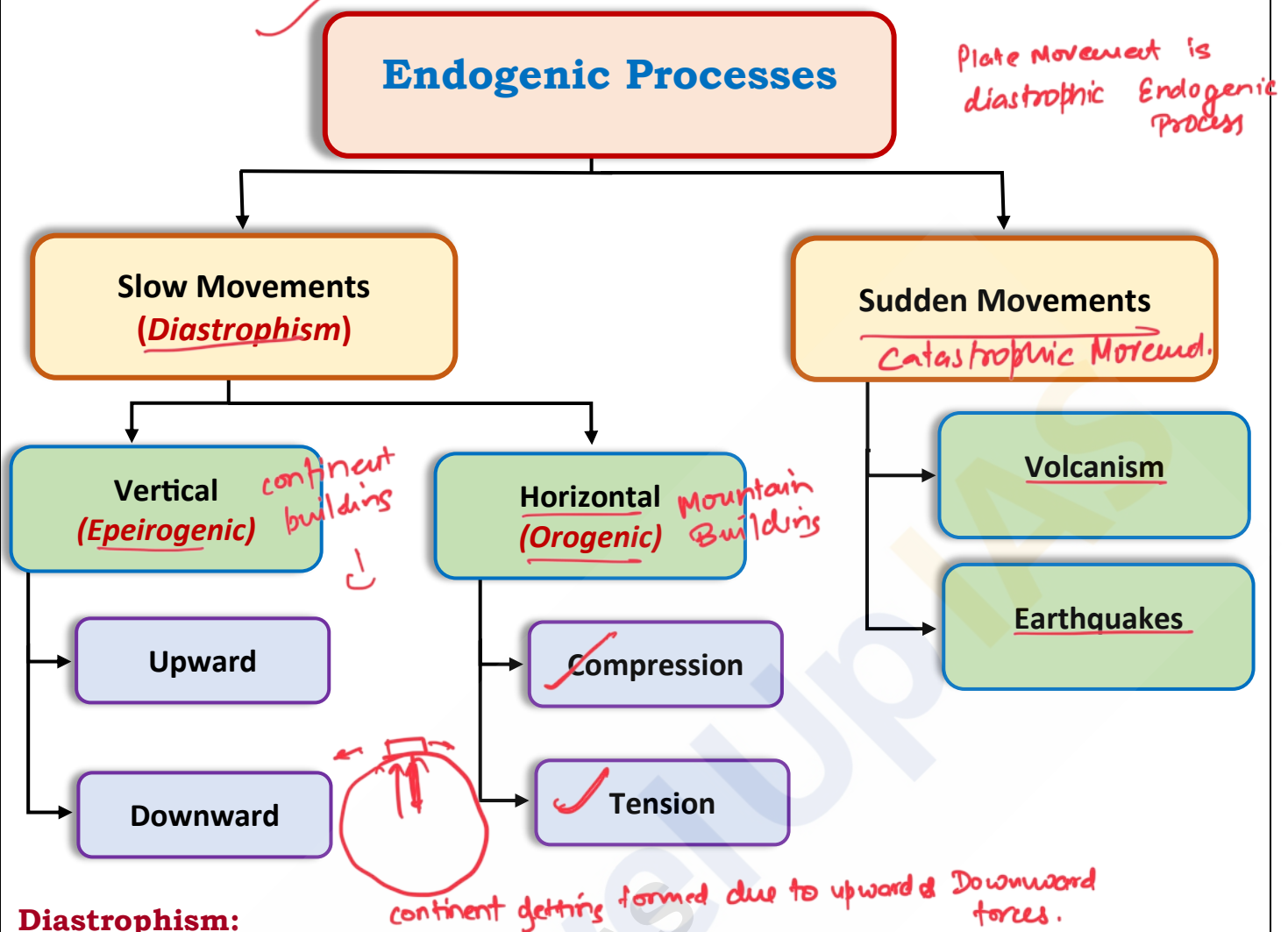
Do You Know?

These forces slow down with time.

- Endogenic Processes cause **two types of movements**:

- Horizontal Movements:** Side to side movements of the Earth's crust, causes lot of disruption as they involve **compression and tension** of the pre-existing rocks.
- Vertical Movements:** Originate from the center of the Earth causing large scale **upliftment or subsidence** of a part of the Earth's crust forming continents and plateaus.

Classification of Endogenic Processes:



Diastrophism:

Diastrophism refers to all processes that **move, elevate, or deform the earth's crust** due to **diastrophic forces** which include **folding, faulting, warping and fracturing**. It can further be classified into **Epeirogenic movements** and **Orogenic movements**.

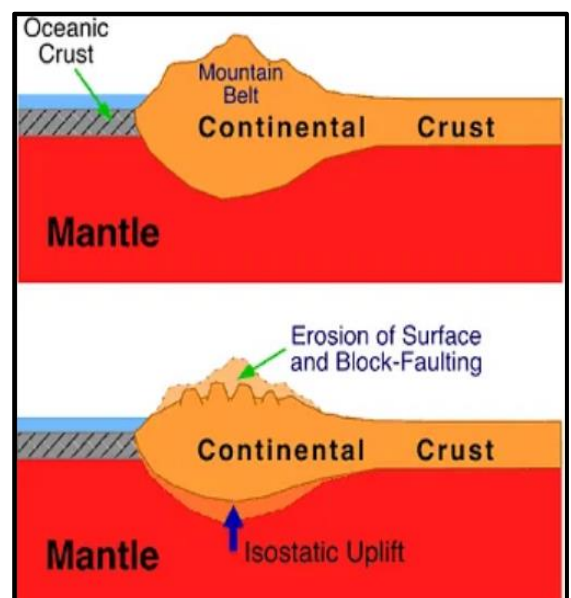
- **Epeirogenic or Continent building movements:** These are vertical movements of the Earth's crust that can cause the **uplift or subsidence** of large areas of land. It acts **along the radius** of the earth therefore; they are also called **radial movements**. Their direction may be towards (**subsidence**) or away (**uplift**) from the center.

- **Uplift** - Raised beaches, elevated wave-cut terraces, sea caves and fossiliferous beds above sea level are evidences of uplift.

- **Example:** Some of the beaches which have been elevated (uplifted) as much as 15 m to 30 m above the present sea level, such as **Kathiawar, Nellore, and Thirunelveli coasts**.

- **Subsidence** - Submerged forests, valleys as well as buildings are evidence of subsidence.

- **Example:** An earthquake (1819) submerged a part of **Rann of Kachchh**. Also, the Andamans and Nicobars have been isolated from the Arakan coast by the submergence of the intervening land.

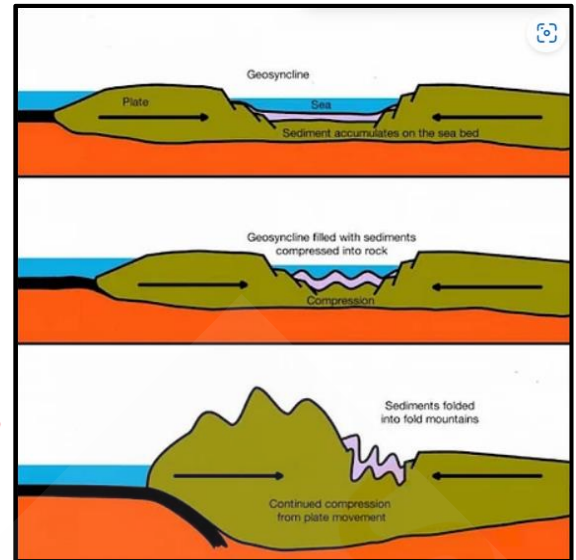


due to compression

- **Orogenic or Mountain building movements:** They are driven by the **interaction of tectonic plates (Earth's crust)** acting tangentially to the Earth's surface. It can further be divided into **Compressional** and **Tensional** forces.

- **Compressional forces** - It results into folding, crustal bending, local rise and subsidence because this type of force acts towards a point from one or more direction.
- **Tensional forces** - It creates cracks, ruptures, features and faults, since this type of forces acts away from a point in two directions.

Block Mountain



Sudden Movements:

These are the result of **long period preparation deep within the Earth**. Only their cumulative effects on the Earth's surface are quick and sudden. It occurs mostly at the lithospheric plate margins. Geologically, these sudden forces are termed as '**constructive forces**' as it creates certain relief features on the Earth's surface. It can be of **two types**: **Earthquakes** and **Volcanoes**. (***) Already done in previous classes)

Structures produced by Endogenic Forces:

Endogenic forces create deformation in the Earth's crust and forms various structures as discussed below:

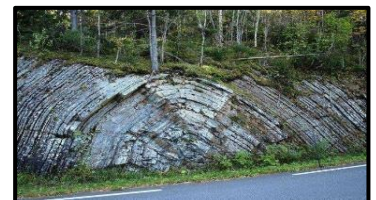
Folds:

Folds are bends in the Earth's crust owing to **compressional forces**. Folds can be formed by compressional forces, such as the collision of tectonic plates, or by extensional forces, such as the separation of tectonic plates.

Types of Folds:

- **Symmetrical Fold:**

- These are ordinary folds. The limbs of the folds are equally inclined on either side. Their axial plane is vertical.



- **Asymmetrical Fold:**

- One of the limbs is more inclined than the other. Their axial plane is inclined.



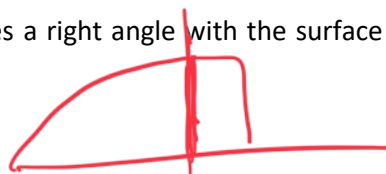
- **Monoclinal Fold:**

- In this fold, one limb makes a right angle with the surface but the other limb is ordinarily inclined.



- **Isoclinal Fold:**

- The two limbs are so much inclined in such a way that they appear equally inclined and parallel to each other.



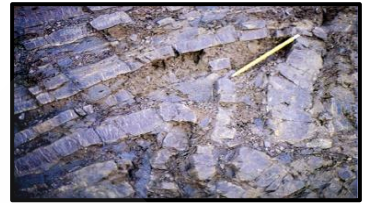
- **Recumbent Fold:**

- In this fold the two limbs are so much inclined that they become horizontal.



- **Overturned Fold:**

- Here one limb is overturned over the other limb. The difference between the overturned and recumbent folds is that the overturned limbs are not horizontal like those of recumbent fold.



- **Plunge Fold:**

- If the axis of the fold is not parallel to the horizontal but makes an angle with it, it is known as Plunging Fold.



- **Fan Fold:**

- It is a great anticline which has many small anticlines and synclines. It is also known as Anticlinorium. A great syncline having many small anticlines and synclines is called Synclinorium.



- **Open Fold:**

- If the angle between the limbs of a fold is obtuse, the fold is called Open fold.



- **Closed Fold:**

- If the angle between the limbs of a fold is acute, the fold is called Closed fold.



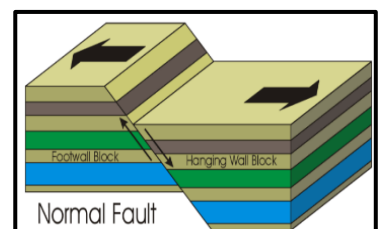
Faults:

A fault is a fracture or zone of fractures in the Earth's crust where the **rocks on either side have slipped relative to each other**. This displacement occurs due to both **tensional and compressional** forces acting horizontally or vertically or both together at the same point of time.

Types of Faults:

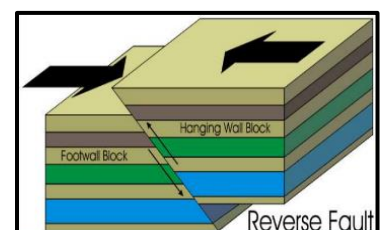
- **Normal Fault (Tension):**

- Forms when **two blocks of the Earth's crust are pulled apart** due to **tensional forces**. The hanging wall (the block above the fault plane) moves down relative to the footwall (the block below the fault plane). It is generally associated with **divergent plate boundaries**.



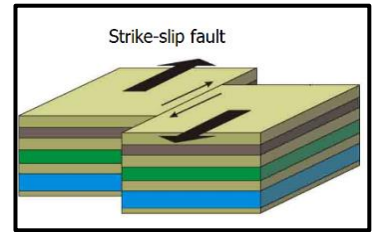
- **Reverse Fault (Compression):**

- Created by **compressional forces that push two blocks** of the Earth's crust together. In this case the hanging wall moves up relative to the footwall. It is often associated with convergent plate boundaries and **mountain-building processes**.



Strike-Slip Fault:

- Occurs due to **horizontal movement along the fault line** caused by **shearing forces**. The two blocks move past each other laterally in opposite directions. Common at transform plate boundaries like the **San Andreas Fault** in California.

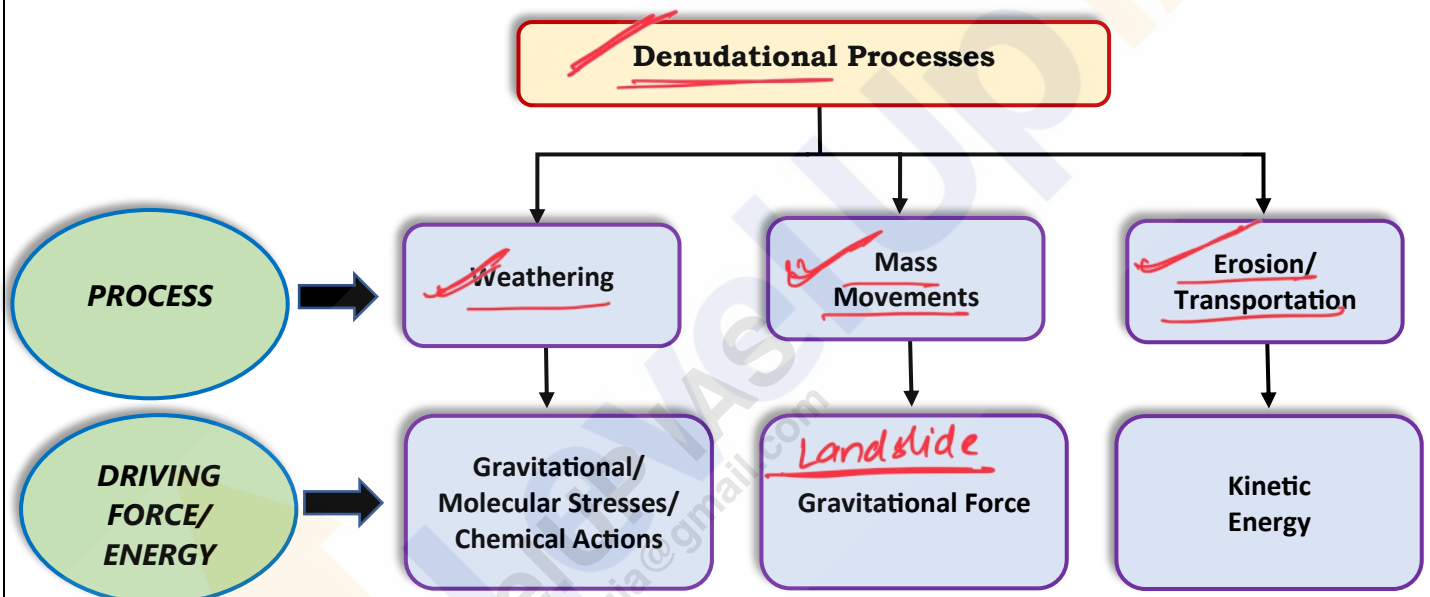


Exogenic Processes:

Exogenic processes refer to the forces and processes that **operate on the Earth's surface**, responsible for reshaping the land and modifying its features through external mechanisms. It derives their energy from atmosphere determined by the **prime source, the Sun**.

These processes work on the Earth's surface and are primarily responsible for the **erosion, transportation, and deposition of materials**. **Temperature** and **precipitation** are the two important climatic elements that control various processes by inducing **stress** in earth materials.

All the exogenic processes are covered under general term **Denudation**. Denudation includes **weathering, mass wasting, erosion and transportation**.



For each process there exist a distinct driving force or energy. As there are different climatic regions on the Earth's surface the exogenic processes vary from region to region. The density, type and distribution of vegetation which largely depend upon precipitation and temperature exert influence indirectly on exogenic geomorphic processes.

Exogenic Processes also depend upon differences in wind velocities and directions, amount and kind of precipitation and its intensity, the relation between precipitation and evaporation, daily range of temperature, freezing and thawing frequency, depth of frost penetration etc. **The sole driving force behind all the exogenic process is the Sun.**

When climatic factors are common the intensity of action depend on type and structure of rocks. Structure includes folds, faults, orientation, inclination of beds, presence or absence of joints, bedding planes hardness, softness of constituent minerals, chemical susceptibility of mineral constituents, the permeability or impermeability.

Different types of rocks offer varying resistances to various geomorphic processes. **The effects of exogenic forces may be small and slow but in long run they have greater effects.** Hence the surface of the Earth is operated by different geomorphic processes and at varying rates.

Weathering:

Weathering is action of elements of weather and climate over Earth materials. Weathering is defined as **mechanical disintegration and chemical decomposition of rocks through the actions of various elements of weather and climate**. There are a number of processes within weathering which act either individually or together to affect the Earth materials in order to reduce them to fragmental state. Weathering processes are conditioned by many complex geological, climatic, topographic and vegetative factors.

As very little or no motion of materials takes place in weathering. It is an **Insitu** disintegration or breakdown of rock material. It decreases the strength of the rocks but **increases permeability and porosity of the rock**.

There are three major groups of Weathering Processes:

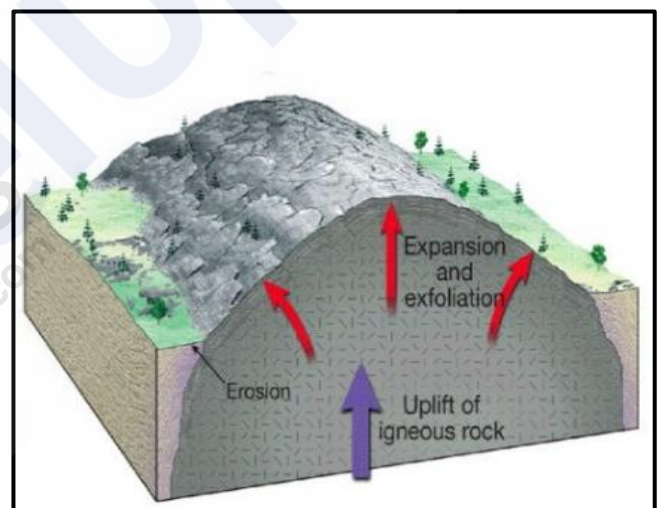
1. **Physical or Mechanical Weathering.**
2. **Chemical Weathering.**
3. **Biological Weathering.**

Physical or Mechanical Weathering:

Physical weathering, also known as **mechanical weathering**, is the process of **breaking down rocks and minerals into smaller pieces without changing their chemical composition**. This transformation of rocks into fragment particle occurs through mechanical methods such as **freezing, thawing, salt weathering, thermal cracking etc**. Physical weathering can be of several types:

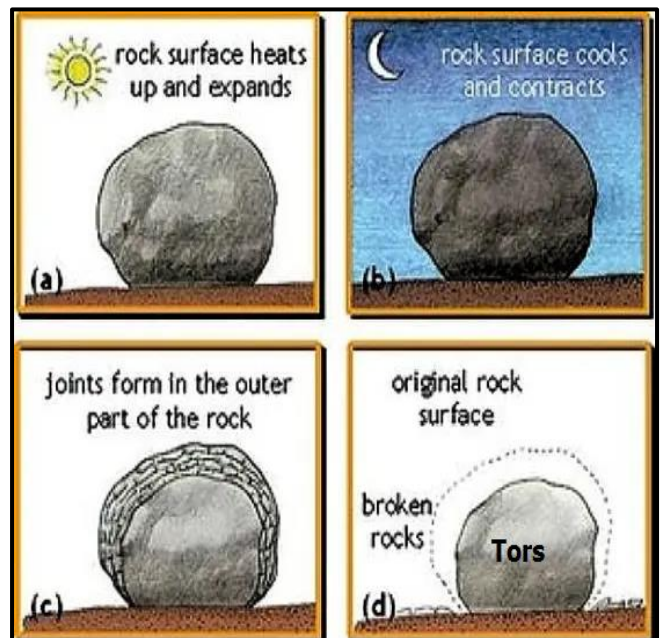
Unloading and Expansion:

- Intrusive igneous rocks formed deep within the Earth's crust are subjected to immense pressure from the overlying layers. As **erosion gradually removes the top load, it leads to a release of vertical pressure**. Consequently, the uppermost layers of the **rock expand and develop fractures parallel to the surface**. Over time, these fractures cause sheets of rock to detach from the exposed surface, a process known as **"Exfoliation or Sheetting"**.



Temperature Change and Expansion:

- Also known as **Thermal stress weathering**. Occurs when rocks are subjected to significant temperature fluctuations. It's particularly prominent in regions of **dry climates** and **high elevations** where diurnal (daily) or seasonal temperature variations are rapid. The process involves the **expansion and contraction** of rock materials due to temperature changes, resulting in stress within the rocks and thus ruptures the rock over the long term. It also creates exfoliation in rounded rocks. This exfoliation in rocks, like **granite**, creates round boulders with smooth surface called **'Tors'**.

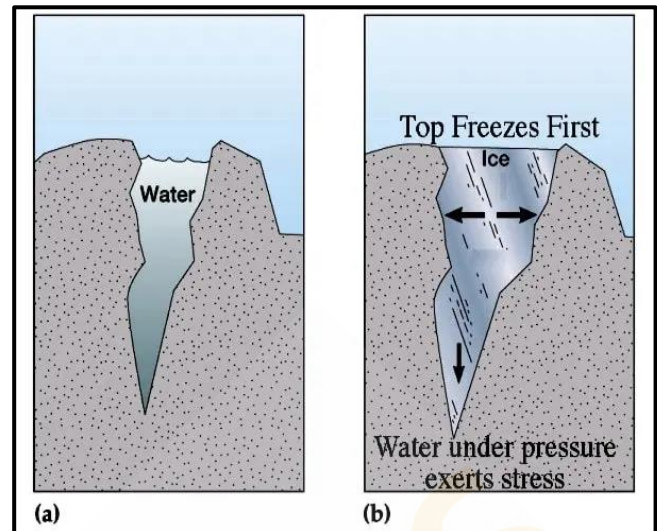


Freezing, Thawing and Frost Wedging:

- When water seeps into cracks or pores in rocks, **freezes, and expands during colder temperatures**, exerting pressure on the rock. These cycles of freezing and thawing cause the rock to weaken and break down over time, a process known as '**Frost wedging**', contributing to the fragmentation of the rock into smaller pieces. It is found in **colder regions** where water freezes and thaws repeatedly.

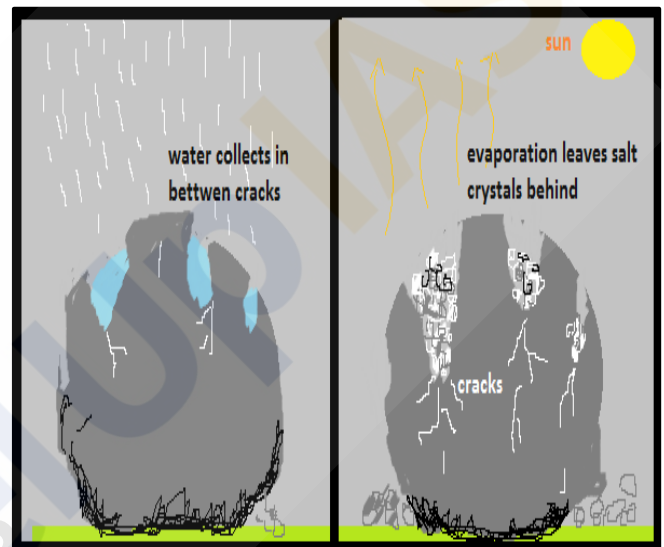
Do You Know?

The volume of water expands as much as by 9% when it freezes.



Salt Weathering (Haloclasty):

- Salt weathering is most common in **coastal areas and in deserts**, where salt is abundant. It occurs when saline water infiltrate rock crevices or pores, subsequently crystallizing and expanding as the water evaporates. This expansion creates internal pressure, leading to the fracturing and breakdown of the rock. Salt weathering can also cause rocks to exfoliate.



Chemical Weathering:

Chemical weathering is the process by which rocks and minerals are broken down and decomposed by **chemical reactions**. It is caused by chemicals such as **water, acids, and oxygen**. Chemical weathering processes include **dissolution, solution, carbonation, hydration, oxidation and reduction** that act on the rocks to decompose, dissolve or reduce them to a fine state. Chemical weathering can be of several types:

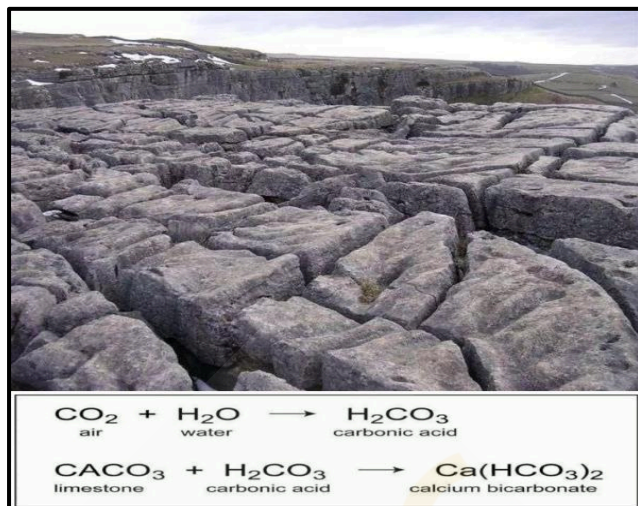
Solution:

- Solution is the result of a **substance dissolving in water or acids**, creating a liquid mixture. The formation of a solution happens when solids dissolve due to **water or ion action** in water. It relies on the mineral's solubility in water or weak acids. Minerals like nitrates, sulphates, potassium, and others prone to dissolution are easily washed away in rainy climates and concentrate in arid regions. For instance, minerals such as **calcium carbonate and calcium magnesium bicarbonate in limestone dissolve in water with carbonic acid** and are carried away. Rock-forming minerals like common salt (sodium chloride) are similarly affected by this process of solution.



Carbonation:

- Carbonation involves the **reaction of carbonate and bicarbonate ions** with minerals, notably feldspars and carbonate minerals. This process is highly active in environments rich in carbon dioxide. **When carbon dioxide combines with water, it forms carbonic acid, a key factor in the breakdown of mineral surfaces.** The dissolution of calcium and magnesium carbonates in carbonic acid leads to the formation of caves, demonstrating the impact of carbonation. The presence of **atmospheric carbon dioxide, when dissolved in rainwater, induces acidity**, initiating the carbonation process.



Hydrolysis:

- Hydrolysis occurs when **mineral ions react with water ions (OH and H)**, leading to the **decomposition of the rock surface**. New compounds are formed, increasing the solution's pH through the release of hydroxide ions. It is particularly efficient in weathering **common silicate and alumino-silicate minerals** due to their electrically charged crystal surfaces.



Oxidation and Reduction:

- Oxidation occurs between **minerals and oxygen leads to the formation of oxides and hydroxides**. It removes electrons from minerals, **making their structure less rigid and more unstable**. This reaction commonly results in red and yellow staining in soils, particularly in tropical regions **with high temperatures and rainfall, caused by iron and aluminium oxides**. Reduction occurs in environments lacking oxygen, typically below the water table or in waterlogged areas, causing the red colour of oxidized iron to turn greenish or bluish grey.



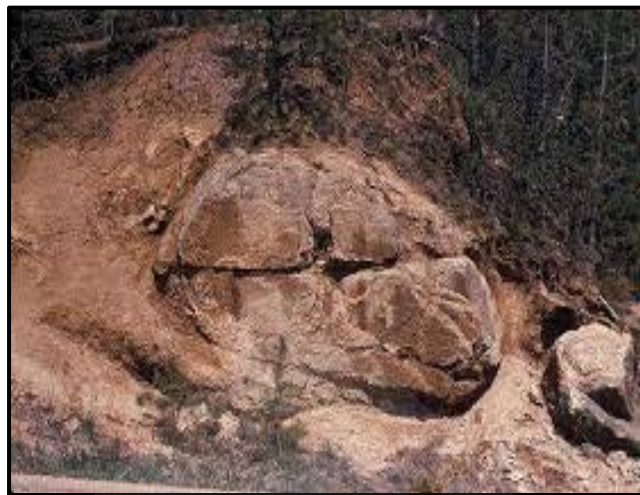
Chelation:

- In this process, the organic **chelating agent** which has **high affinity for metals, binds the metals** thus decomposing the rocks. (Humus breaking down to **humic acid** is example of chelation.)



Hydration:

- In hydration, **water molecules attach themselves firmly to minerals or compounds, resulting in an expansion of the mineral volume and rendering it less stable.** For instance, **calcium sulphate absorbs water and transforms into gypsum**, which is less stable. This process is **both reversible and gradual**. Repeated cycles of hydration can induce fatigue in rocks, potentially leading to their disintegration. The volume changes caused by hydration can also contribute to physical weathering through processes like **exfoliation and granular disintegration**. Furthermore, hydration can facilitate other decomposition reactions by expanding the crystal lattice, providing a larger surface area for further reactions to occur.



Do You Know?

These all-chemical weathering processes are inter-related to each other and go hand in hand and hasten the weathering process.

Biological Weathering:

It is the weakening and subsequent disintegration of rock by **plants, animals, and microbes**. It is contributed to or removal of ions and minerals from the weathering environment and physical variations due to movement or development of organisms. Key biological weathering processes include:

- Root Wedging** - Plant roots growing into rock fractures can **exert pressure, causing the rock to crack and break apart**. As roots expand, they widen existing cracks, leading to rock disintegration.
- Biological Activity** - Burrowing animals, such as **rodents and insects**, **contribute to weathering by creating tunnels and fissures in rocks**, which can lead to rock fragmentation and soil formation.
- Lichen and Moss** - These organisms **secrete weak organic acids that break down minerals in the rock**, contributing to its disintegration. Their growth can also physically weaken the rock surface.
- Decay** - The **decomposition of organic matter**, including dead plants and animals, **contributes organic acids to the environment**, accelerating the weathering of surrounding rocks.



Significance of Weathering:

- **Soil formation:** Weathering is essential for soil formation. As **rocks and minerals are weathered, they break down into smaller and smaller pieces**, which eventually form soil. Soil is important for plant growth and for supporting ecosystems.
- **Rock and mineral breakdown:** Weathering breaks down rocks and minerals into smaller pieces, making them easier to **transport by water, wind, and ice**. This process helps to create new landforms, such as mountains, valleys, and deltas.
Weather rock less surface Runoff more percolation
- **Groundwater recharge:** Weathering helps to **recharge groundwater supplies**. As rainwater infiltrates the ground, it dissolves minerals and other substances, which can then be taken up by plants or used by humans.
- **Landform Creation:** Weathering processes, alongside erosion, **play a pivotal role in the creation of various landforms**, including valleys, canyons, caves, and coastlines, contributing to the diversity of landscapes.
- **Contribution to the Carbon Cycle:** Chemical weathering **absorbs carbon dioxide** from the atmosphere, converting it into carbonate minerals, which are eventually deposited and stored in the form of sedimentary rocks.
- **Mineral release:** Weathering **releases minerals into the environment**. These minerals can be used by plants and animals, or they can be used to make products such as fertilizer and cement.
- **Economic resources:** Weathering can help to form valuable economic resources, such as **mineral deposits and bauxite** (the ore from which aluminium is extracted).
- **Environmental Balance:** Weathering processes are part of the **Earth's natural recycling mechanisms**, maintaining a balance in the Earth's geosphere, hydrosphere, and atmosphere.
- **Human Impact:** Understanding weathering processes is crucial in fields such as **agriculture, construction, and environmental conservation**. It influences soil quality, building materials, and landscape management.
- **Aids in Geological Studies:** By observing weathering patterns, geologists gain insights into **past environmental conditions**, helping in the interpretation of Earth's history.

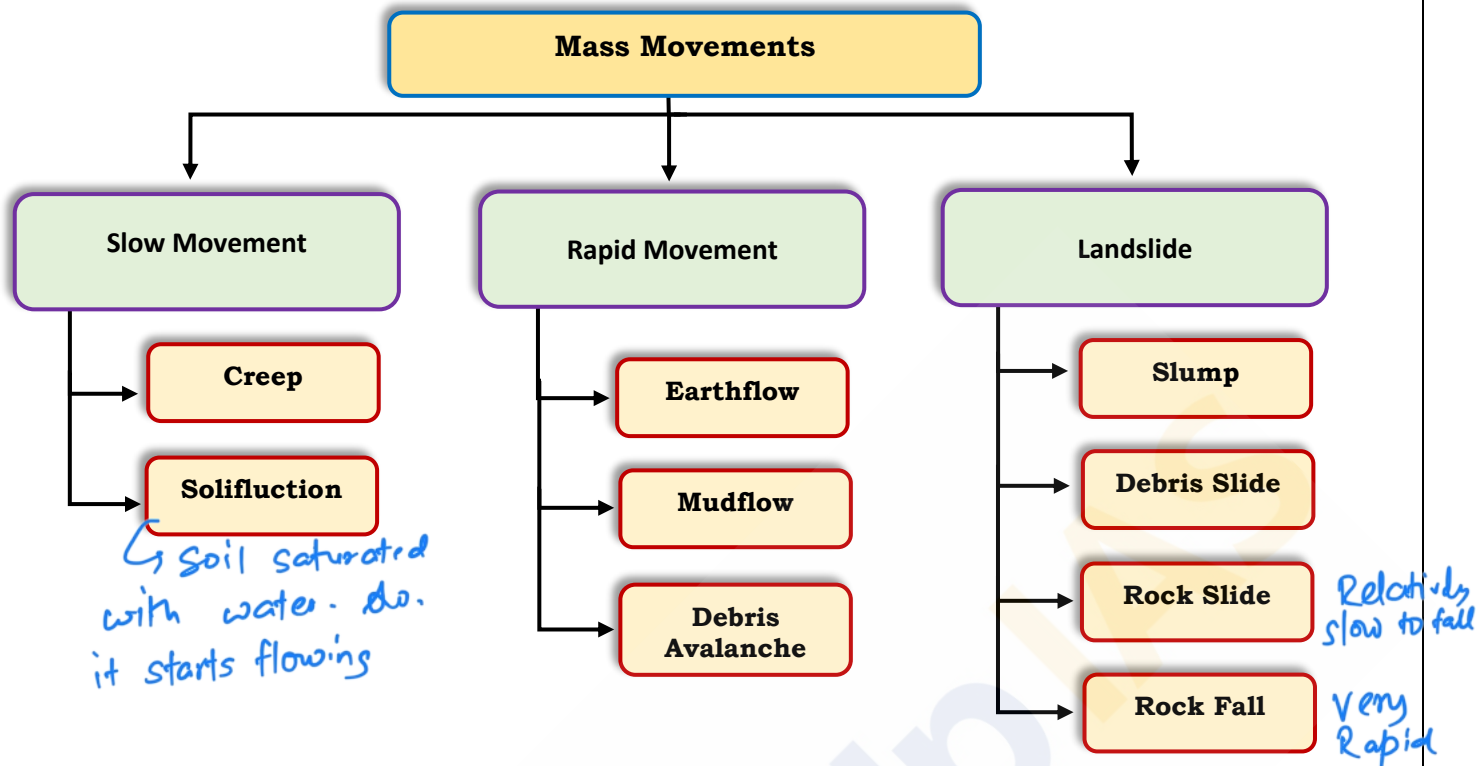
Mass Movements:

Mass movement, often referred to as **Mass wasting**, is the process by which rock and debris are transferred down the slopes under the **direct influence of gravity**. This movement can occur at varying speeds, be it slow or fast, and it is primarily driven by the force of gravity. Unlike processes like erosion, which involve the actions of agents like water, glaciers, wind, or waves, mass movement is solely a product of gravity. **While weathering may aid in mass movements, it is not a prerequisite for them.**

Key factors are responsible for Mass movements:

- **Gravity** - Primary force pulling materials downslope, more pronounced on steeper slopes.
- **Water** - Increases material weight and can act as a lubricant on slopes.
- **Rock and soil type** - Loose, unconsolidated materials are more prone to movement.
- **Vegetation** - Root systems help stabilize, while deforestation raises movement risk.
- **Human activity** - Construction, mining, and deforestation can disturb slopes, increasing vulnerability to mass movement.

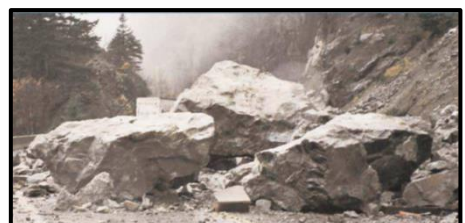
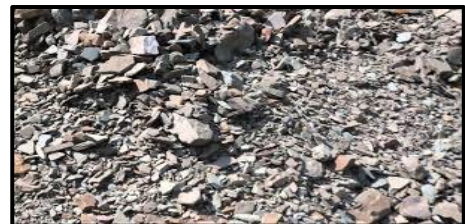
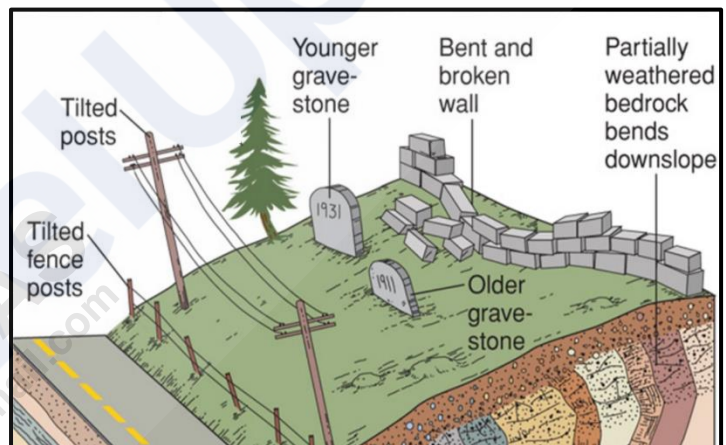
Types of Mass movements:



Slow Movements:

Creep:

- Creep is the **slow, gradual, and imperceptible movement** of soil down a slope due to the **force of gravity**. This movement is often facilitated by processes such as **daily thawing and freezing or grazing**, which loosen the soil and allow gravity to take effect.
- There are different types of creeps based on the material involved:
 - Soil Creep** - This involves the **slow movement of loose soil** under the influence of **gravity**.
 - Talus Creep** - Also known as scree, it is the **gradual downslope movement of rock fragments**.
 - Rock Creep** - This refers to the **intermittent slipping movement of rocks down a slope**.



Rapid Movements:

Rapid movements are most **common in humid climatic regions** and occur over a range of slopes, from gentle to steep. Various types of rapid movements include:

Earth Flow:

- Earth flow involves the movement of **water-saturated clayey or silty earth materials down low-angle terraces or hillsides**. On steeper slopes, even bedrock, particularly soft sedimentary rocks like shale or deeply weathered igneous rocks, may slide downslope.



Mudflow:

- Mudflows occur when thick layers of weathered materials, in the absence of vegetation cover and **under heavy rainfall, become saturated with water and flow down along definite channels**. This flow resembles a stream of mud within a valley. When mudflows emerge from channels onto piedmonts or plains, they can be highly destructive, engulfing roads, bridges, and houses. Mudflows **frequently happen on the slopes of erupting or recently erupted volcanoes**.



Debris Avalanche:

- Debris avalanches are more characteristic of humid regions, with or without vegetation cover, and **occur in narrow tracks on steep slopes**. These avalanches can be **much faster than mudflows** and are **similar to snow avalanches**.



Landslides:

- Landslides involve relatively **rapid and noticeable movements of dry materials**. The size and shape of the detached mass **depend on the nature of discontinuities in the rock, the degree of weathering, and the steepness of the slope**. Several types of movements are identified in this category:
 - **Slump** - The **slipping of one or several units of rock debris** with a **backward rotation** concerning the slope.
 - **Debris Slide** - Rapid rolling or sliding of earth debris **without backward rotation** of the mass.
 - **Debris Fall** - Nearly a **free fall of earth debris** from a vertical or overhanging face.
 - **Rockslide** - Sliding of individual rock masses **on bedding, joint, or fault surfaces**. Over steep slopes, rock sliding can be very fast and destructive, occurring as planar failures along discontinuities like steeply dipping bedding planes.
 - **Rockfall** - The free falling of rock blocks over any steep slope, **keeping themselves away from the slope**.

